- The nuclear material and cytoplasm of the male gametocyte divides to produce long, actively motile, whip – like forms of 8 microgametes. This process is called exflagellation of male gametocytes.
- The Exflagellation is completed within 15–30 minutes for *P. falciparum*.
- The female gametocyte does not divide but maturation involves by condensation of nucleus to become the female gamete.
- Female gamete is fertilized by one of the microgametes to produce the zygote. The zygote is formed in 20–120 minutes after the blood meal. The zygote is initially is a non – motile round body, but within 18–24 hours, it gradually elongates into a vermicular motile form. This is called the ookinete.
- Ookinete penetrates the epithelial lining of stomach wall. Their anterior end comes in close contact to the cell membrane by secretion of some proteolytic substances which causes lysis of cell membrane. Later, the ookinete come to lie just beneath the basement membrane.
- It becomes rounded into a sphere with an elastic membrane. This stage is called the oocyst. The oocyst increase in size and undergo numerous nuclear multiplication which develops a large number of sickle shaped bodies known as sporozoites.
- Number of oocysts in the stomach wall varies from a few to over a hundred.
- Around the 10$^{th}$ day of infection the oocyst ruptures, releasing sporozoites in the body cavity of the mosquitos.
- The sporozoites are distributed through the circulating fluid into various organs and tissues of the mosquito except the ovaries.
- The sporozoites have a special affinity towards the salivary glands. The mosquito at this stage is capable of transmitting infection to man.

### 8.5.7 Pathogenesis

In malaria, typical pathological changes are seen primarily in the spleen, liver, bone marrow, lungs, kidney and brain.

**Liver:** The liver is enlarged. The organ becomes more firm and pigmented. Pigments are found in parenchymal cells.

**Spleen:** The spleen is markedly enlarged. If the infection lasts over a long period, the spleen is usually grayish, dark brown or even black and is commonly known as 'ague cake'.

Bone marrow, Lungs, Kidneys and Brain are enlarged and pigmented. They are filled with parasitized erythrocytes. Anemia is caused by destruction of large number of red cells by complement mediated and autoimmune hemolysis. It is also due to the increased clearance of both parasitied and parasitized RBCs by the spleen.

### 8.5.8 Clinical Manifestations

The incubation period is generally 9–14 days but, it can be as short as 7 days. The most malignant form of malaria is caused by P. *falciparum* hence, variable clinical syndromes are associated with falciparum malaria. That include,

1. Prodromal (initial indication of the onset of disease) period: Non – specific symptoms such as malaise (condition of



TN_GOVT_XII_Micro_Biology_CH08.indd 175 27-02-2019 14:53:05